

Rana Dolui

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SUMMARY

I am Rana Dolui, a full-stack developer specializing in high-performance backend systems, scalable APIs, and AI-driven applications. Experienced in Go, Python, and TypeScript, with strong expertise in distributed systems and cloud infrastructure.

SKILLS

- **Languages:** JavaScript, TypeScript, Python, Golang
- **Frameworks & Libraries:** Next.js, React, Express.js, FastAPI, Flask, Django, Fiber
- **Databases & Baas:** PostgreSQL, MongoDB, SQLite, Firebase, Prisma, Drizzle, Redis, Convex, Supabase
- **DevOps & Cloud:** Docker, Kubernetes, Argocd, Helm, AWS, CI/CD, Kafka, Cloudflare, Rabbitmq

WORK EXPERIENCE

React Native Developer | HumanEra Intelligence | Intern SEP 2024 - DEC 2024

At HumanEra Intelligence, I developed a social media-type app using React Native and Expo. I followed a local database-first approach and explored techniques to keep the app alive in the background, implementing the necessary permissions for both Android and iOS. I also contributed to the backend, which was built with Express.js, GraphQL, and Prisma

Full stack Developer | Saciva | Full Time JAN 2025 - NOV 2025

At Saciva, I develop scalable web applications and design optimized APIs with efficient database management. I build responsive UIs and implement real-time web notifications. I developed a recommendation system using Typesense and a custom classification model. I also design microservices and manage deployments using GitHub Actions and AWS (EC2, S3, RDS).

Python Developer | Telo | Part time DEC 2025 - Feb 2026

At Telo, I handle backend development and AI agent systems, primarily using FastAPI, Redis, and PostgreSQL. I build AI-driven features with LangChain and OpenAI to enable automation and intelligent workflows. I also manage deployments on VPS using Docker and GitHub Actions, ensuring system stability and reliability.

Freelancing | Part time NOV 2025 - PRESENT

I work as a freelance backend developer, building and fixing backend systems using FastAPI, Fiber, Express.js, and Hono.js. I handle large projects, performance issues, and secure API design, and I'm comfortable working with messy codebases. I also work with PostgreSQL, Redis, Docker, and Kubernetes, managing everything from development to deployment on VPS or cloud.

OPEN SOURCE PROJECTS

FlashORM: High-Performance Multi-Database ORM (Go)

- A high-performance ORM in Go with sqlc-like functionality, focused on speed and type safety.
- Designed type-safe code generation supporting Go, JavaScript, TypeScript, and Python.
- Achieved up to 64× faster performance than Prisma and 15× faster than Drizzle through optimized query execution.
- Implemented support for multiple databases: MySQL, PostgreSQL, SQLite, MongoDB, and Redis.
- Developed a plugin-based architecture and built-in Studio for schema management, enabling database operations with or without using the ORM.

AstraAPI: High-Performance Backend Framework for Python (C++)

- Built a high-performance backend framework for Python, implemented in C++ for maximum speed and efficiency.
- Developed a custom native HTTP server, eliminating ASGI/Starlette dependencies with a fully C++-based runtime.
- Designed a lightweight multi-worker architecture, **using ~100 MB RAM for 3 workers (vs. ~300–500 MB in typical Python setups)**.
- Achieved **84–88% worker efficiency**, optimizing concurrency and resource utilization.
- Engineered the system to **handle ~200K requests/second per worker**, enabling high scalability and throughput.

PROJECTS

- **OmniAI:** Developed a full-stack AI platform where users can chat with any type of file or website via URL. Supports document uploads, OCR, and contextual Q&A using semantic search. Built with Next.js, Go/Fiber, FastAPI, PostgreSQL, Redis, Pinecone, Docker, gRPC, RabbitMQ, and LangChain
- **AiResume:** AI-powered resume SaaS for generating resumes and cover letters, analyzing CVs, and producing ATS-friendly results. Includes an admin panel, user and plan management, Razorpay payments, and full role-based access control. Built with Next.js, Firebase, OpenAI, FastAPI, Razorpay, MySQL, Docker, and Nginx.
- **Proxyon** Developed a website that allows users to configure an API gateway or load balancer by connecting nodes (like n8n), enabling setup with just a few clicks. Built with Next.js, React Flow, TailwindCSS, Framer Motion, Go, Fiber, NATS, MongoDB, PostgreSQL, Docker, Nginx, S3, Terraform, and CoreDNS.
- **PeerDrop:** is a secure file-sharing tool that enables real-time, peer-to-peer transfers with end-to-end encryption. Developed using React, TypeScript, Go, and Gorilla WebSockets, it ensures fast, private, and seamless file sharing without relying on third-party servers.
- **HireLens:** is an AI-powered platform that helps users improve their interview skills. Built with Next.js, Prisma, TypeScript, and Gemini, it offers personalized mock interviews, real-time feedback, and expert insights, making interview preparation more effective and confidence-boosting.
- **khelratna:** A WebView-based application for hosting cricket tournaments and local matches. It includes an admin panel to manage users, tournament owners, captains, and more, with a robust role-based access control system. Built with Next.js, Socket.IO, Go, FastAPI, Supabase, Flutter, Redis, Docker, and Firebase.
- **Passman:** An ultra-secure password manager with end-to-end zero-knowledge encryption. It supports authenticator app-based verification, live session management, and a zero-trust mode that allows only a single active, IP-bound session to prevent session hijacking. Built with Svelte, FastAPI, Redis and PostgreSQL.

EDUCATION

Tech in Computer Science Engineering –

Bengal Institute of Technology

September 2023 – July 2027
